

Solution Requirements

Date	29 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely log in using email and password to access personalized crop recommendations.</i>	High
2	Authorization levels	<i>Admin manages datasets and models. Farmers can access crop prediction, weather information, fertilizer, and irrigation recommendations.</i>	High
3	External interfaces	<i>Integrates with OpenWeatherMap API for live weather data and GitHub for project management.</i>	High

4	Transactions processing	<i>Accepts soil nutrient and weather inputs, processes them using the ML model, and generates crop recommendations instantly.</i>	High
5	Reporting	<i>Displays crop recommendations, weather conditions, fertilizer suggestions, irrigation advice, and prediction graphs.</i>	Medium
6	Business rules, etc.	<i>Crop recommendations are generated only after validating user inputs. Weather data is updated in real time before prediction.</i>	High
7	Compliance to laws or regulations	<i>Protect user information and follow general data privacy guidelines.</i>	Medium
8	<i>Other</i>	<i>Maintain prediction history for future analysis and continuous model improvement.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The application should generate crop recommendations quickly after user input.</i>	<i>Response time less than 3 seconds</i>
2	Scalability	<i>The system should support an increasing number of users and larger agricultural datasets.</i>	<i>Supports 1000+ concurrent users</i>
3	Security & Data Privacy	<i>Protect user data through secure communication and proper authentication mechanisms.</i>	<i>Secure login and encrypted communication</i>
4	Reliability & Availability	<i>The application should remain available with minimal downtime.</i>	<i>99.9% availability</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy for farmers to use on different devices.</i>	<i>Responsive design with mobile compatibility</i>
6	<i>Other</i>	<i>The project should follow modular coding practices for easy maintenance and future upgrades.</i>	Modular architecture and documented code